critical porosity, shear deformation can proceed with no further volume change. It is extremely difficult to make packings looser than the critical porosity, and such packings tend to be metastable under loading. Experiments have been reported by Roscoe, Schofield and Wroth[3], who deformed an assemblage of 1-mm steel balls in simple shear and found that the critical porosity was approximately 39 per cent. This value is slightly dependent on the stress-level, due, presumably, to the elasticity of the balls.

If the deformation of granular media is interpreted from the point of view of thermodynamics, the relationship between the dilatancy and the geometry of packing becomes clear. For a unit volume, the first law of thermodynamics can be written as[4]:

$$\frac{\mathrm{d}u}{\mathrm{d}t}=\sigma_{ij}\dot{\varepsilon}_{ij}-q_{i,i} \tag{1}$$

It is clear that the internal energy of an aggregate of rigid, rough balls is zero, and if the deformation is sufficiently slow to be isothermal, then all the work being done on the system is dissipated as heat. Therefore:

$$\sigma_{ij}\dot{\varepsilon}_{ij}=q_{i,i} \tag{2}$$

For an irreversible process, the change in entropy per unit volume may be expressed by:

$$T\frac{\mathrm{d}s}{\mathrm{d}t}=T\frac{\mathrm{d}s_e}{\mathrm{d}t}+T\frac{\mathrm{d}s_i}{\mathrm{d}t} \tag{3}$$

where s_e is the entropy flux and s_i is the internal entropy production, which must be greater than or equal to zero.

The entropy flux is defined by:

$$T\frac{\mathrm{d}s_e}{\mathrm{d}t}=-q_{i,i} \tag{4}$$

From equations (2), (3), (and 4):

$$\sigma_{ij}\dot{\varepsilon}_{ij}=T\frac{\mathrm{d}s_i}{\mathrm{d}t}-T\frac{\mathrm{d}s}{\mathrm{d}t}$$

$$\text{or}\quad s_i\dot{e}_{ij}+3p\dot{\varepsilon}_v=T\frac{\mathrm{d}s_i}{\mathrm{d}t}-T\frac{\mathrm{d}s}{\mathrm{d}t} \tag{5}$$

If it is assumed that:

$$T\frac{\mathrm{d}s_i}{\mathrm{d}t}=3p\dot{\varepsilon}_v \tag{6}$$

then when the assembly is deformed with no further irreversible volume change, there is no further internal entropy production and the entropy is a maximum. For an assemblage of single-sized, rigid, balls with Coulomb friction acting at the contacts, only configurational entropy obtains. Hence the porosity computed from the geometry of random packing of equal spheres should be equal to that observed at the critical porosity. The porosity of the model described by Bernal[2] is 39 per cent and within experimental limitations equals that observed by Roscoe, Schofield and Wroth[3]. It can be concluded that granular materials undergoing plastic deformation with no further volume change assume the most random configuration which corresponds to the state of maximum entropy.

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GEOPHYSICS

Magnetization of Basalt of Bed I, Olduvai Gorge, Tanganyika

Palæomagnetic data which bear on the time-scale of geomagnetic polarity epochs are here reported for a radiometrically dated Pleistocene basalt flow in East Africa. The occurrences in various places of stratigraphic groups of rocks which are magnetized either with the sense of the present geomagnetic field (normal polarity) or with the opposite sense (reversed polarity) have been interpreted to mean that the geomagnetic field has reversed itself at various times in the geological past[1,2]. The accumulated evidence for the reality of such field reversals has recently been reviewed[3,4], and palæomagnetic investigations of radiometrically dated volcanic rocks have permitted a preliminary attempt by Cox *et al.* to bracket the ages of the most recent reversals[5].

During the course of geological mapping of the Pleistocene succession at Olduvai Gorge, Tanganyika[6], four accurately oriented hand specimens were collected from the uppermost 8 ft. of the basalt member of Bed I (as redefined by Hay[6]). The sampling location is the south side of Olduvai Main Gorge, about 50 yards west of the junction of the Main and Side Gorges, at lat. 2° 58′ 45″ S., long. 35° 20′ 30″ E. The basalt here is olivine-rich and consists of thin lenticular flow units, each having a maximum thickness of 2–4 ft. Each sample was collected from a different flow unit; three from highly vesicular parts of flow units and one from relatively non-vesicular and coarser-grained lava. Perfect preservation of the surface pahoehoe crusts and absence of interbedded material indicate that the flow units were extruded over a short period of time, not more than a few tens of years. For palæomagnetic purposes, all the samples will be considered to represent a single point in time.

Three 1-in. cores were cut from each hand-specimen, and their directions and intensities of magnetization were measured with a 'spinner' magnetometer. The average intensity of magnetization is $5{\cdot}5 \times 10^{-3}$ e.m.u., with little variation. The directions are shown in Fig. 1*a*, in which it is seen that all specimens have normal polarity but that there is a slight streaking toward the present geomagnetic field direction. The three cores from the coarse-grained specimen show the most deflexion; this is apparently a viscous magnetization induced by the present field. In order to remove any secondary components and to test for stability of magnetization, ten of the cores were partially demagnetized by tumbling about three axes simultaneously in an alternating field of 100 oersteds peak intensity. The two remaining specimens were partially demagnetized by heating to 250° C in nitrogen and cooling in field-free space. Directions of magnetization after 'cleaning' are shown in Fig. 1*b*. The remaining streaking is in declination only and probably results from compass needle deflexion during specimen orientation. Mean directions of magnetization of all cores both before and after demagnetization are given

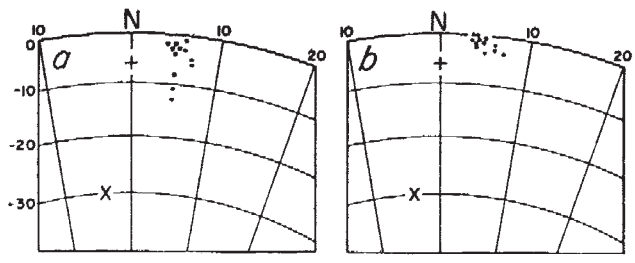


Fig. 1. Equal-area projections of directions of magnetization of basalt from Olduvai Gorge: (*a*) original measurements, (*b*) after partial demagnetization. +, Axial dipole field; ×, present field; ●, north-seeking directions, upper hemisphere

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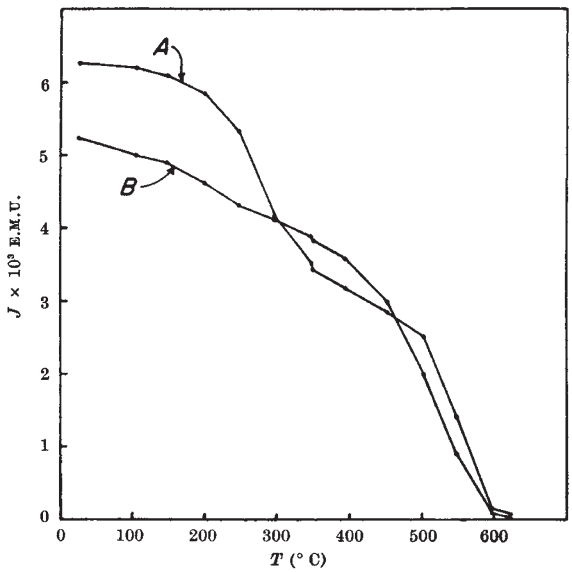


Fig. 2. Thermal decay of intensity J of natural remanent magnetization of (A) vesicular and (B) coarse-grained basalt from Olduvai Gorge. (Measurements made at room temperature)

in Table 1, together with Fisher's[7] statistical parameters.

The opaque minerals of one of the highly vesicular specimens are seen in polished section to consist of skeletal crystals of pink titaniferous magnetite containing bright patches and lamellæ, also isotropic, which are probably due to oxidation. The more coarse-grained specimen contains a much more heterogeneous opaque assemblage consisting chiefly of pink titaniferous magnetite grains, which have been partly or completely oxidized to titanomaghemite, to magnetite–ilmenite intergrowths, or to ilmenite–hæmatite pseudomorphs. Some primary ilmenite–hæmatite intergrowths also occur in this specimen.

Thermal demagnetization of the natural remanences of these two specimens is illustrated in Fig. 2. Curve A corresponds to the vesicular basalt; the main magnetization appears to reside in two minerals with Curie temperatures about 325° C and 590° C. There is also a minor component with curie temperature above 595° C which is probably due to traces of hæmatite. The magnetization of the coarse-grained specimen (curve B) decreases more monotonously, in keeping with the more heterogeneous composition of its oxides; but again there is a vestige of magnetization remaining after heating to 595° C. In both specimens the directions of all components of natural thermo-remanent magnetization are parallel and have the same sense, and the directions remain stable up to 595° C. Although spontaneous self-reversal subsequent to cooling in a reversed field cannot be entirely ruled out for these rocks[8,9], it appears highly unlikely.

The age of this basalt has recently been redetermined by the potassium–argon method and found to be approximately 1·8 million years (J. F. Evernden and G. H. Curtis, quoted by Hay[6]). The oldest reversely magnetized rock discussed by Cox *et al.* is 1·69 ± 0·10 million years old; the youngest of the older group of normally magnetized rocks shown in their Fig. 1 is about 2·4 million years old. The two tentative geomagnetic polarity chronologies proposed by Cox *et al.* represent apparent periodicities of either 0·5 or 1 million years; the present result is consistent with both. In terms of the latter scheme, which is the simplest and which Cox *et al.* favour, the data from Olduvai Gorge combined with their data conservatively restrict the time of the next-to-most recent polarity reversal (the N2-R1 boundary) to between 1·9 and 1·6 million years.

Table 1. Average North-seeking Directions of Magnetization

	D*	I†	N	R	k	α_{95}
Original measurements	5·4°	5·1°	12	11·975	400	1·9°
Partially demagnetized	4·8°	1·6°	12	11·995	2,200	0·9°

* Declination east of north.
† Inclination upward.

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Discordant Lead–Uranium Ages due to Continuous Loss of Lead

Since the existence of certain regularities in discordant lead–uranium ages was pointed out by Ahrens[1,2] considerable interest has been shown in determining the cause or causes for the discordance.

The interpretations which have been offered so far fall into two categories. Episodic loss of lead (or gain of parent uranium) has been considered by Wetherill[3,4], who showed that a linear discordant lead–uranium age pattern on a $^{206}Pb/^{238}U$ versus $^{207}Pb/^{235}U$ diagram may be explained by a single episode of lead loss. The possibility of continuous loss of lead, on the other hand, has been suggested by Russell and Ahrens[5], who noted that the apparent time of lead loss appears to be a function of the age of the minerals. Nicolaysen[6] has considered solid diffusion of lead atoms as a possible cause of age discordance and has developed graphical procedures to determine the true age of the mineral and the diffusion coefficient which has governed the loss of lead.

A very important contribution has recently been made by Tilton[7], who has also commented on the remarkable coincidence in the agreement in the apparent time of lead loss for old minerals from all over the world. By assuming that discordant ages are due to loss of lead from mineral spheres by continuous volume diffusion, Tilton has shown that for an expected range in the diffusion coefficient, the diffusion model produces an identical age-discordance pattern to the episodic loss model. The important modification is, however, that the time of abstraction of lead has now no significance.

It is the purpose of this communication to show that the linear relationships observed are not specific to episodic and diffusion loss mechanisms, but occur for many loss processes.

The equation of radioactive decay modified to include the effect of loss of daughter may be written:

$$\frac{dD}{dt} = \lambda P - \Lambda D$$

Where P and D are the concentrations of parent and daughter isotopes, λ is the decay constant and Λ the loss of daughter.

Assuming zero loss or gain of the parent and a constant loss of daughter, the solution of the foregoing equation becomes:

$$\frac{D}{P} = \frac{\lambda}{\lambda - \Lambda}[e^{(\lambda-\Lambda)t} - 1]$$

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